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Data Fusion Between Land and Sea: Multi-Isotope Fingerprints of Viking Animals and Modern Plants

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***Abstract*—Bioarchaeology aims to reconstruct, e.g., the diet or provenance of animals and humans in archaeological times. This can be done by investigating so-called multi-isotope fingerprints, resulting from the analysis of several different isotope systems in parallel. However, the multi-isotope fingerprint of samples from coastal regions can be influenced by the so-called sea spray effect, resulting in "too marine" isotope signatures in terrestrial herbivorous individuals, falling in-between less or un-affected herbivores and marine mammals when clustering the isotope data ("sea spray cluster"), what cannot be explained by the diet or habitat of these individuals. The recently proposed in-between instance (IBI) definition allows the detection of additional sea spray candidates, not grouped into the sea spray cluster, thus less affected by sea spray but still of interest to domain experts. The sea spray effect locally expected in archaeological individuals can also be investigated by isotope analysis in modern plants of the same region. The fusion of isotope data measured in archaeological bones and in modern plants allows us to investigate samples (un-)affected by sea spray, what is relevant for domain experts to understand the potential local range of isotope values at an archaeological site.**

***Index Terms*—cluster analysis, data fusion, domain expert knowledge, GMM, in-between instance**

I. Introduction

A. Data Mining and Bioarchaeology

Multi-isotope data is an important source for understanding ancient animal and human diets, habitats, and migration patterns [1]. Data sparsity in bioarchaeology requires the use of data mining to effectively manage and interpret data [2]. Clustering is a data mining technique for organizing sparse data, revealing hidden patterns, and extracting knowledge [3] and has been shown to be successful in bioarchaeology [4].

Data points differing from the usual pattern are identified as outliers or anomalies [5]. Outlier detection is also important for highlighting unique or unexpected findings in the data, which are especially interesting from an archaeological perspective [6], [7]. The recent proposal of the **"in-between instance" (IBI) concept** [8] adds a new dimension to data analysis, suggesting that these instances may provide unique insights in bioarchaeological research. An object is considered to be an IBI if it is not a member of any cluster, but is located *between* two or more clusters. The semantic behind the term *between* depends on the chosen predicate, e.g., neighborhood, class probability, or geometric criteria. The motivation for detecting such patterns is that IBIs may potentially exhibit characteristics of the clusters they are in between.

B. Isotope Analysis on Archaeological and Modern Samples

The analysis of **stable isotopes** (e.g., ^{13}C) is an established tool in bioarchaeology [1]. While radioactive isotopes, such as ^{14}C, decay by emitting radioactive radiation, stable isotopes remain unchanged over time and, therefore, provide stable markers for archaeological material. When regarding heavy elements, the quotient of heavy and light isotope is given (e.g., ^{87}Sr/^{86}Sr). For light elements (e.g., C, N, O, S), the isotope ratio is compared with a standard material, presented in delta notation (in ‰; e.g., δ^{13}C) [9].

$\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$ allow the investigation of the protein source in the diet of an archaeological individual. Further insights on the diet and habitat features are reflected by $\delta^{13}C_{carb}$, which can, e.g., be used to distinguish individuals living in different habitats. Oxygen isotope values ($\delta^{18}O_{carb}$) reflect diet (metabolic water sources) and provide information on the habitat and provenance of an individual [9].

Coastal regions are potentially affected by sea spray aerosols, containing ions (e.g., sulfate) of marine origin with a distinct isotopic signature. The sea spray salts are taken up by plants, which are further consumed by animals. Consequently, both plants and animals (including humans) from coastal regions can show a seemingly marine isotope signal in different isotope systems, not explainable by their diet, especially in the case of terrestrial herbivores (see, e.g., [10], [11]). This so-called **sea spray effect** was shown to be present in $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $\delta^{34}S_{coll}$, and ^{87}Sr/^{86}Sr values measured in archaeological skeletal remains [10]–[12] and can also be detected in (leaf) cellulose ($\delta^{13}C_{cell}$, $\delta^{18}O_{cell}$), total sulfur ($\delta^{34}S_{total\ S}$), and strontium isotope ratios (^{87}Sr/^{86}Sr) measured in plant leaves [13], [14]. Without a more detailed understanding of the sea spray effect, diet and provenance reconstructions are falsified and, thus, limited at coastal sites. Advancing the clustering of in-between instances can help identify individuals potentially affected by sea spray.

Conventional one- or two-dimensional isotope data analysis is insufficient in identifying patterns present in the isotope data [15]. Due to the variable information provided by the different isotopic systems, a multi-isotope approach is necessary [2]. (Multi-)Isotope data can be quite complex, especially in coastal environments, due to the high number of potential parameters, including sea spray, that affect the different isotopic systems in different ways. Related work shows that

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machine learning approaches are required to allow a sufficient interpretation of these **multi-isotope fingerprints** by domain experts [2], [4], [15]–[17].

We aim to study 1) if the existence of in-between characteristics can be combined with domain knowledge and 2) what insights are drawn from this observation, compared to, e.g., clustering. However, in order to investigate the complex sea spray effect on isotope data, a novel approach is required. Archaeological samples usually lack specific information on, e.g., habitat or provenance, besides the information reflected in their isotopic signature, which can, however, be (highly) misleading due to the sea spray effect. On the other hand, isotope data of modern environmental samples, such as plants, from known sites can be investigated, allowing to identify the potential regional sea spray impact. However, we have to consider changes over time when comparing those results with isotope data of archaeological remains. By combining archaeological and modern isotope datasets and machine learning techniques, we address the following research questions:

1) Are archaeological individuals seemingly (highly) affected by sea spray identifiable as IBIs?
2) Can we fuse modern environmental and archaeological bone isotope data in order to identify the locally expected sea spray magnitude at a site or in a region?

II. Material

The study region is located in Northern Germany, close to the Baltic Sea. Archaeological remains were excavated at the important Viking Age trading center **Haithabu** (Hedeby; 804-1066 AD) and its medieval successor **Schleswig** (11th-14th cent. AD), located at the Schlei inlet close to the Baltic Sea (Fig. 1). Haithabu and Schleswig played a key role as **supra-regional trading centers** over a period of more than five centuries (9th-13th cent. AD) [18]. Thus, several animals excavated at both sites were probably imported [19].



Fig. 1. Map of northern Germany and southern Denmark showing the sample sites.

This study focuses on four isotope datasets:

Dataset - Archaeological Bones (A) (n = 87) contains four features, representing different stable isotope systems ($\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$, $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$) measured on archaeological bone samples from Haithabu (HB) and Schleswig (SL) of terrestrial herbivorous mammals (n = 61) and marine mammals (n = 26) [12]. Prior work on this dataset has shown the presence of the sea spray effect on some of the herbivores [12].

Dataset - Sulfur (AS) (n = 22) is a subdataset of dataset A, additionally including sulfur isotope data ($\delta^{34}S_{coll}$) [12].

Dataset - Plant (P) (n = 57) is part of a pilot study on the investigation of the sea spray effect in modern environmental samples [14] with four features, namely the isotope signatures ($\delta^{13}C_{cell}$, $\delta^{18}O_{cell}$, $\delta^{34}S_{total\ S}$, $^{87}Sr/^{86}Sr$) of modern plant leaves collected at 19 different sites along the Baltic coast, with sample sites around the archaeological sites of Haithabu (HB) and Schleswig (SL), at Oehe-Maasholm (OE), along the coasts of Fehmarn island (FE), nearby Rødby (RO) at the southern coast of Danish Lolland, and around Rerik (RE) as shown in Fig. 1 (see [14] for details).

Dataset - West (PW) (n = 17) is a subdataset of dataset P, only including plants collected close to the archaeological sites of Haithabu and Schleswig; thus, plants collected close to Haithabu, Schleswig, or Oehe-Maasholm (western region in Fig. 1). For comparison with datasets A and AS, the measured $\delta^{13}C_{cell}$ and $\delta^{34}S_{total\ S}$ values are calculated into $\delta^{13}C_{carb}$ ($\delta^{13}C_{carb} = \delta^{13}C_{cell} + 14.1‰$; [20]), as expected for ruminant consumers capable to digest plant cellulose, and $\delta^{34}S_{coll}$ values ($\delta^{34}S_{coll} = \delta^{34}S_{total\ S} + 0.5‰$; [11]), as expected for herbivorous consumers of plants indicating the respective isotopic signature (see [14] for details). Both approximated measures can be seen as valid predictors for the isotopic signature of herbivorous consumers of the same region.

As described above, the datasets differ with respect to the investigated material (bone vs. plant), the dating of the samples (archaeological vs. modern), the region(s) of origin, the number of dimensions (4 vs. 5 dimensions), and the isotope systems included in the datasets. The **heterogeneity of the data** is a challenge for the data fusion approach in particular. We have to consider differences in the isotope signature when comparing samples from **different regions** (e.g. Schleswig vs. Fehmarn) **and time periods** (archaeological vs. modern).

III. Theory and Methods

A. Sea Spray and In-Between Instances

In order to discover IBIs and individuals affected by sea spray, we propose a 3-step approach:

Step 1: Lower Marine Limit Threshold

Definition 1. *Given a one-dimensional isotope system δ with n samples $x_i \in \mathbb{R}$ where each sample is assigned a label $y_i \in \{H, M\}$ with terrestrial herbivorous mammals H and marine mammals M . Because of their different diets and habitats, H and M should fall into two distinct, non-overlapping groups when clustering their isotopic signatures. Furthermore, a **lower marine limit** m is defined. This lower marine limit is defined for each isotope system as the marine piscivorous mammal, local at the (archaeological) site under study, exhibiting the lowest measured isotope value (see [12]). A sample $x_i \in H$ is labeled as "too marine" (TM) and considered as potentially affected by sea spray with respect*

to its delta-value if $x_i > m$ holds, resulting in an overlap of H and M as depicted in Fig. 2. If an overlap is observed for several isotope systems, the confidence that a particular sample was actually influenced by sea spray increases.

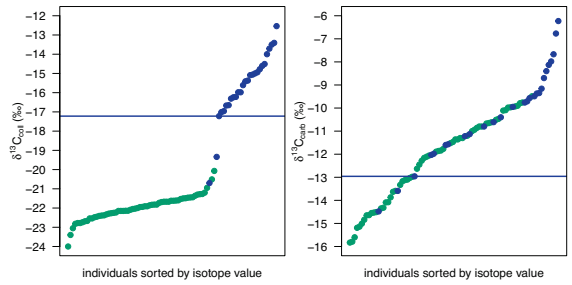


Fig. 2. Isotope values (Dataset A) investigated according to Definition 1. $x_i \in H$ depicted in green, $x_i \in M$ depicted in blue. The lower marine limit m (defined according to [12]) is shown as horizontal line.

By applying this threshold we can identify potential sea spray samples, however we cannot make any statements yet on potential IBIs. For this we pursue further analysis in the upcoming step.

Step 2: IBI Detection In order to detect IBIs, we have to rely on the following assumptions:

- Due to the distinction between H and M, we observe a partitioning of the data into two clusters resulting from a Gaussian Mixture Model (GMM), hence we also have the z-scores for each sample available
- The two groups do not overlap in *at least* one dimension (see Definition 1), hence reflecting no sea spray impact but a data structure capable of distinguishing H and M

For the IBI detection, we rely on the following semantic:

Definition 2. *Given n samples $x_i \in \mathbb{R}^D$, with $\mathbb{R}^D$ resembling a set of isotope systems $\{\delta_1, ..., \delta_D\}$. For a subset of samples it holds that their labels are $y_i \in \{H, M\}$. Cluster means μ_H and μ_M (obtained from Gaussian Mixture Model clustering; see III-B) are denoted for both classes H and M. A sample x_i is considered an IBI if the following predicate is satisfied:*

$$dist(x_i, \mu_M) \leq dist(\mu_M, \mu_H) \wedge dist(x_i, \mu_H) \leq dist(\mu_M, \mu_H)$$

Definition 2 can be applied to full-dimensional settings ($\mathbb{R}^D$), however, there is a certain property that raises the need to determine if an object is an IBI based on their one-dimensional representation (single isotope system):

Definition 3 (IBI Dimensionality-Property). *A sample x_i that is identified as IBI in full-dimensional space $\mathbb{R}^D$ implies that there exists at least one x_i in a lower-dimensional space $\mathbb{R}^d$, with $d < D$ identified as an IBI:*

$$y_i \in \{IBI\} : x_i \in \mathbb{R}^D \Rightarrow y_i \in \{IBI\} : x_i \in \mathbb{R}^d$$

In this use case, it holds that an IBI can be labeled as TM (according to Definition 1); however, not every TM is also defined as an IBI. Hence, we need to discover IBIs on single isotope systems as follows:

Definition 4 (Dimension-wise IBI Detection). *Given the same conditions as in Definition 2, a sample x_i is considered as an IBI if the following predicate is satisfied:*

$$\bigvee_{\delta=1}^{D} \Big(dist(x_i[\delta], \mu_M[\delta]) \leq dist(\mu_M[\delta], \mu_H[\delta]) \\ \wedge dist(x_i[\delta], \mu_H[\delta]) \leq dist(\mu_M[\delta], \mu_H[\delta]) \Big) = 1$$

Definition 4 states that an object is considered an IBI if the predicate holds in at least one feature $[\delta]$.

Individuals potentially affected by sea spray (TM) and IBIs have been identified at this stage. What remains unaddressed is the *magnitude or certainty* of in-betweenness or sea spray impact. For this purpose, the set of potential sea spray samples TM from *Step 1* are unified with the detected potential IBI from *Step 2*: $\Lambda = TM \cup IBI$. For determining the sea spray magnitude, we compute the z-scores (see below) of the samples in Λ. Both IBIs and individuals highly affected by sea spray should show lower clustering probabilities than expected for non-overlapping clusters.

Step 3: 3-cluster case If a sea spray effect is present in at least one of the examined isotopic systems, one would expect a third cluster, besides H and M, namely a "mixed cluster" or "sea spray cluster" (S), falling in between the others. Samples within H are probably unaffected by sea spray or affected to a smaller degree, while samples grouped into S should have been markedly affected by sea spray. In cluster S, we can expect both marine mammals to exhibit a (unusual) high isotopic freshwater signal and terrestrial herbivores with a seemingly marine isotopic signal, both a result of the coastal environment (see [12] for details).

At this point, the question arises: which of the detected samples in Λ coincide with S? Specifically, which samples in Λ are not captured by the sea spray cluster S? In more formal terms, we are particularly looking for samples from the subset $\Omega = \Lambda \setminus S$. Do we, e.g., observe $\Omega \approx IBI$?

B. Cluster Analysis

Gaussian Mixture Model (GMM) clustering is a powerful probabilistic clustering approach. The GMM model represents a population of data as a mixture of multivariate Gaussian distributions. The chosen algorithm (EM) iteratively optimizes the GMM model, resulting in ellipsoidal clusters with means μ_k. The geometric properties of the clusters (shape, volume, and orientation) are described by covariance matrices Σ_k.

As a soft clustering approach, GMM clustering determines the probability z_k for each data point belonging to each of the k clusters ($\sum_{i=0}^{k} z_k = 1$) and assigns data to the cluster with the highest z-score. Besides the information on clustering quality, a low z-score could be of particular interest with respect to IBI

and sea spray impact.GMM clustering was performed using the R package mclust [21],

GMM clustering has been successfully applied in different isotope studies [2], [12] and was shown to result in valid output even for comparably small isotope datasets [15].

C. Outlier Detection

Isolation Forest (IF) is an unsupervised ensemble method that we used to detect outliers [22], i.e., catch the sea spray individuals in our datasets. IF aims to find anomalous individuals that are fewer in number and are more isolated compared to normal data points, making them easier to separate, i.e., isolate. IF is based on the collection of multiple isolation trees (here: n = 88). Anomaly detection occurs over two phases as outlined in [23].

D. Data Fusion

The archaeological bone and modern plant isotope datasets (see II) include different isotope systems due to the different sample materials (bone vs. plant leaves). Datasets A/AS and Dataset P do not share any isotope system. As a consequence, these isotope datasets cannot be directly fused. However, plant Dataset PW, including approximated $\delta^{13}C_{carb}$ and $\delta^{34}S_{coll}$ values (see II), and the (archaeological) Dataset AS share two isotope systems and can, thus, be fused.

IV. Results and Discussion

A. Marine Threshold

According to the definition of the **lower marine limit** (Definition 1) a number of herbivores of the four-dimensional Dataset A can be identified as (probably) affected by the sea spray in either $\delta^{13}C_{carb}$ (n = 35, 57%) or $\delta^{18}O_{carb}$ (n = 61, 100%) or both (n = 35, 57%). No overlap and, thus, no sea spray is observable in both $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$, reflecting the dietary protein unaffected by sea spray [12].

B. IBI and Clustering Probability

GMM clustering on dataset A with a given number of two clusters, including a cluster of terrestrial herbivorous mammals (cluster 0) and marine mammals (cluster 1), respectively, reveals some individuals with comparably low clustering probability (Fig. 3). The majority of these individuals fall in-between the two clusters. Two herbivores are misclassified and sorted into the marine cluster with a clustering probability of almost 1. Both were detected as outliers using IF.

A total of 78 individuals (88.6%) of the dataset were defined as **IBI in at least one dimension**. Since sea spray is only present in the group of herbivores (H), we focus on the IBIs within this class (n = 58, 65.9%). In addition, based on domain expert knowledge, we can rule out any sea spray effect in both $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$. Excluding IBIs defined in only $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$, respectively, or two-dimensional IBIs of these two features, gives 36 IBIs (40.9%). Λ includes a total number of 81 individuals. Considering again only herbivores gives 61 entries for Λ. None of these individuals is defined as an IBI in only $\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$, or both.

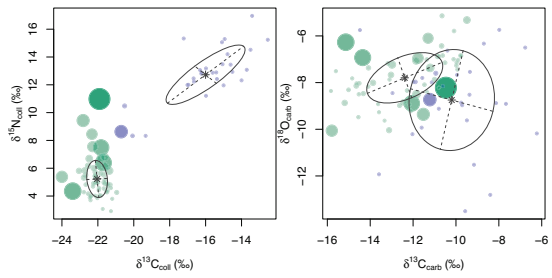


Fig. 3. GMM clustering results for Dataset A with two clusters (cluster 0: green (H), cluster 1: blue (M)), also indicating clustering uncertainty (large filled symbols: >95% quantile, smaller open symbols: 75-95% quantile, small dots: <75% quantile).

Clustering probability is generally high for all individuals of both cluster 0 (0.9977±0.0061) and 1 (0.9997±0.00184). However, nine individuals, falling in-between the clusters - including a marine mammal and eight herbivorous mammals - show slightly lower z-scores between about 0.965 and 0.994 (Fig. 3). All these individuals were also defined as IBI and might show an especially high sea spray signal in at least one isotope system.

C. Sea Spray Cluster

Several terrestrial individuals in the dataset are affected by sea spray, resulting in an overlap of class H and M due to terrestrial herbivores exhibiting a "too marine" isotopic signature which is (partly) more marine than that of local marine mammals (IV-A; see [12] for details). Accordingly, a third cluster is expected by domain experts, containing a mixed isotope signal resulting from the sea spray effect (Fig. 4). GMM clustering with three clusters revealed that, while individuals of the herbivorous cluster 1 seem to be affected by sea spray to a smaller degree or are even unaffected, herbivores grouped into the "mixed cluster" or **"sea spray cluster"** (cluster 0) are (markedly) affected by sea spray. This sea spray cluster includes two marine mammals exhibiting a higher isotopic freshwater signal or originating from more northern (colder) regions [12], and terrestrial herbivores with a seemingly marine isotopic signal. Six herbivores grouped into the sea spray cluster were defined as outliers based on IF. These individuals show (unusual) high $\delta^{15}N_{coll}$ values, eventually resulting from a freshwater effect in addition to the sea spray, influencing $\delta^{15}N_{coll}$ [12]. Individuals of cluster 2 show a marine protein signal in their $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$, as expected for marine mammals, however, (partly) overlap with the herbivores in the other two isotopic systems due to the sea spray effect [12].

When investigating Ω resulting from the Λ entries as obtained above (IV-B), we find 11 herbivores not included in the sea spray cluster. These individuals were defined as "too marine" in only one dimension ($\delta^{18}O_{carb}$), while no sea spray could be evidenced in $\delta^{13}C_{carb}$. Besides a single individual with a low clustering probability (0.656), all other individuals were grouped into the herbivorous cluster (H) with high

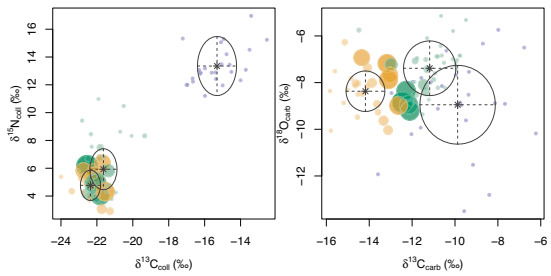


Fig. 4. GMM clustering results for Dataset A with three clusters (cluster 0: orange (S), cluster 1: green (H), cluster 2: blue (M)), also indicating clustering uncertainty (large filled symbols: >95% quantile, smaller open symbols: 75-95% quantile, small dots: <75% quantile).

probabilities close to 1. These individuals likely showed a lower sea spray signal than that observed for individuals of sea spray cluster S. All individuals in Ω show some sea spray signal in their multi-isotope fingerprints, which can, however, not be specified without the IBI approach.

D. Clustering Plant Isotope Data

In contrast to Dataset A, Dataset P does not contain a marine class. All samples in this dataset are terrestrial. Accordingly, we cannot define TM. As visible from Fig. 5, the mixed cluster (cluster 0) in the 3-clusters case includes plants affected by sea spray. However, also plants (probably) unaffected by sea spray are grouped into this cluster. This results from the fact that a group of samples (cluster 1) from four sites show negative sulfur isotope values, reflecting sulfate sourcing from a distinct portion of organic sulfur (see [14] for details). Plants of cluster 2 show a marine isotope signal with respect to, in particular, $\delta^{34}S_{\text{total S}}$ as well as $^{87}Sr/^{86}Sr$ [14], even so only terrestrial plants are included in this study. The sea spray impact on these specimens was particularly high. Clusters also overlap in both $\delta^{13}C_{\text{cell}}$ and $\delta^{18}O_{\text{cell}}$ (Fig. 5). This partly indicates a sea spray impact. However, due to the missing TM label, only the multi-isotope fingerprint currently allows the identification of plants with higher or lower sea spray impact.

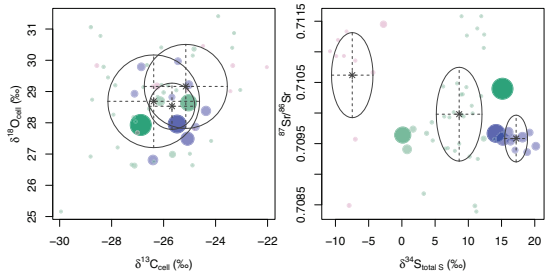


Fig. 5. GMM clustering results for Dataset P with three clusters (cluster 0: green (terrestrial/mixed signal); cluster 1: pink, cluster 2: blue (marine signal)), also indicating clustering uncertainty (large filled symbols: >95% quantile, smaller open symbols: 75-95% quantile, small dots: <75% quantile).

Due to the special characteristics of cluster 2 the mixed cluster (cluster 0) in this case includes not only samples potentially affected by sea spray but also plants most likely unaffected by sea spray. While we can assume a sea spray impact for all plants grouped into cluster 2, the potential sea spray signal is variable in samples of cluster 0. To learn more from such data, we have to include a real marine cluster M. Thus, the fusion of bone and plant isotope data, including terrestrial and marine samples, is performed in the following.

E. Data Fusion

GMM clustering with a given number of three clusters separately applied on the archaeological Dataset S and plant Dataset PW with five and four isotopic systems, respectively, is shown in Fig. 6. As for the larger archaeological Dataset A, a "sea spray cluster" (cluster 0) can be identified in dataset S. All three clusters overlap with respect to $\delta^{34}S_{\text{coll}}$, indicating at least a small impact by marine sulfates in all terrestrial herbivores. Individuals of the first cluster do not show a (distinct) sea spray impact in their $\delta^{13}C_{\text{carb}}$, but in the corresponding $\delta^{18}O_{\text{carb}}$ values (Fig. 6) (see [12] for details).

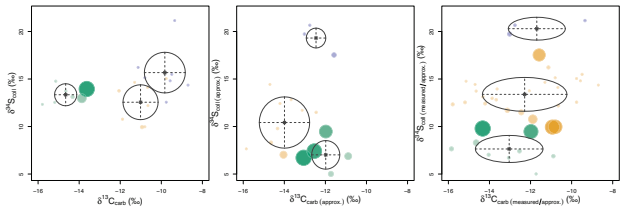


Fig. 6. GMM clustering results for Dataset AS (left), Dataset PW (middle), and a fused dataset AS/PW (right; 2-dimensional; measured and approximated $\delta^{13}C_{\text{carb}}$ and $\delta^{34}S_{\text{coll}}$) with three clusters (cluster 0: orange (S), cluster 1: green (H), cluster 2: blue (M)), also indicating clustering uncertainty (large filled symbols: >95% quantile, smaller open symbols: 75-95% quantile, small dots: <75% quantile).

All plants included in this study are terrestrial, more or less close to the coast, thus exhibiting a potential sea spray signal of varying magnitude [14]. No marine plants (e.g., algae) are included. Accordingly, no real marine cluster (such as for datasets A and AS) is expected. However, cluster 2 identified when performing GMM clustering on the data include coastal plants exhibiting a sulfur isotope value (almost) identical to seawater (see [14]). Cluster 1 contains plants with terrestrial $\delta^{34}S$ values, seemingly unaffected by sea spray (Fig. 6).

When GMM clustering is performed on the **fused data**, containing only $\delta^{13}C_{\text{carb}}$ and $\delta^{34}S_{\text{coll}}$ values - either measured (archaeological samples) or approximated from measured values (plant samples) - the three identified clusters still contain a similar pattern as for the original datasets (Fig. 6). A marine cluster (cluster 2) can be identified, containing plants with an extremely high seawater signal as well as a marine mammal with a distinct seawater isotope signal. Cluster 1 contains only plants, with sulfur isotope signals lower than observed in archaeological individuals but typical for terrestrial environments unaffected by sea spray [11]. Modern plants, as well as archaeological herbivorous and marine mammals, are included in cluster 0. The variance in $\delta^{13}C_{\text{carb}}$ is very high in this cluster, resulting from a marine signature in the marine mammals

and a variable sea spray effect observed in both measured and approximated terrestrial herbivores. From a sulfur isotope perspective, however, most terrestrial individuals included in this dataset show at least a small sea spray signal.

In the case of the fused dataset, 33 (84.6%) samples were defined as IBI, including a total of 28 (71.8%) terrestrial samples (both measured and approximated data). All these individuals are included in Λ. Four samples defined as IBI show low z-scores below 0.80. These individuals were labeled as TM in at least one isotope system, falling in-between clusters H and S (1-dimensional TM) or between clusters S and M (2-dimensional TM). Five (12.8%) terrestrial samples are included in Ω, thus were not captured by the sea spray cluster. While the z-score is low (0.767) for one sample (approximated from a plant sample), also classified as an outlier using IF, the clustering probability was high for the remaining individuals. Besides another (archaeological) sample, which is "too marine" (TM) with respect to $\delta^{13}C_{carb}$ only, the others are identified as TM in both isotopic systems. They were, thus, most likely affected by sea spray, even though they are not grouped into the sea spray cluster.

V. Conclusion

The proposed IBI approach enables detecting additional sea spray candidates, not or only partly identified as "too marine" and not grouped into the sea spray cluster. These individuals, which exhibit a lower but still observable sea spray impact in at least one isotopic system compared to individuals grouped in the "sea spray cluster", are of interest to domain experts but cannot be identified by clustering or IF outlier detection.

This paper also presents the **first attempt to perform cluster analysis on fused isotope data measured in archaeological bones and modern plants** based on sparse pilot data. In a (prehistoric) trading center, such as Haithabu and Schleswig, a high number of non-local individuals has to be expected (e.g., import, migration). Non-local individuals could show a different multi-isotope fingerprint due to, e.g., differences in local food sources and habitats. The variability reflected in the investigated plants, even on a comparably small spatial scale, is high, indicating a varying potential impact by sea spray when consuming plants from even nearby sites. Thus, fusion of measured and approximated isotope values of archaeological bone and modern plant samples is meaningful. The isotope signatures approximated from plant samples are relevant for the re-definition of the terrestrial class unaffected by sea spray, what is not necessarily fully represented by the archaeological samples.

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